

Prutha Kailas Kendre

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Education

Master's in Astro & Particle Physics (M.Sc.)

Eberhard Karls Universität Tübingen, Germany

10/2022 – 12/2025

Tübingen, Germany

Grade: 1.63 Focus: Theoretical frameworks, observational techniques, data analysis methods. Thesis: A Systematic eROSITA Survey for Ultra Cool Dwarfs (UCDs).

Bachelor's in physics (B.Sc.)

Jai Hind College, Mumbai University, India

07/2019 – 04/2022

Mumbai, India

Grade: A+ Focus: Newtonian mechanics, atomic & nuclear physics, mathematical physics, relativity.

Professional Experience

Working Student - Innovation & Economy Team

Cyber Valley, Tübingen, Germany

11/2024 – 12/2025

Tübingen, Germany

- Support in logistics and communication for innovation events and incubator programs.
- Contribution to CRM data organization and cleanup efforts to improve outreach efficiency by 35%.
- Research on tech transfer, AI ecosystems, and startup landscapes in Europe.

Student Research Assistant (HiWi)

University of Stuttgart, Stuttgart, Germany

03/2025 – 09/2025

Stuttgart, Germany

- Developed a Python script to migrate interface functionality from DS9 to the SAMP protocol.
- Writing Python scripts to identify and remove invalid or corrupt files from a dataset of 2+ lakh files.
- Assisting in running and managing over 5 data pipeline modules used in research processing.

Professional Tutor

Hackberry, Mumbai, India

08/2022 – 08/2024

Mumbai, India

- Conducted over 50 online coding sessions for 80+ students in grades 1-7, simplifying complex concepts and fostering a 20% improvement in coding proficiency.
- Leveraged advanced coding and problem-solving techniques applicable to scientific computing and data analysis, enhancing learning outcomes and real-world application skills.
- Taught Python, Web Development, Scratch to 50+ students.

- Delivered 200+ detailed text answers for higher grades across 5+ subjects, maintaining a 98% accuracy rate and ensuring clarity in content.
- Conducted thorough research for various topics to produce high-quality, error-free content, significantly enhancing my research skills and attention to details.

Projects & Publications

Automated Detection of Open Star Clusters (in progress)

- Aim: Develop an automated, unsupervised learning framework capable of detecting and characterizing star clusters.
- Analysis: Designed a density-adaptive, representation-learning-based model that can efficiently identify both known & previously undiscovered clusters. The model focuses on learning latent spatial and kinematic relationships among stars, thereby minimizing human bias and maximizing scalability.
- Expected results: Generalizable AI pipeline for autonomous star cluster discovery and validation.

A Systematic eROSITA Survey for Ultracool Dwarfs (UCDs)

- Aim: Investigate the X-ray emission properties of UCDs using eROSITA data to better understand their role in the stellar population.
- Analysis: Utilized eROSITA X-ray data to identify UCDs and cross-referenced with Gaia data for physical characterization. Analyzed X-ray luminosity and spectral features to explore the relationship between magnetic activity and UCD properties.
- Results: Identified UCDs with significant X-ray emissions, suggesting active magnetic fields. The statistical analysis showed their notable presence in the nearby stellar population, contributing to a better understanding of low-mass star and brown dwarf evolution.

Solar System Formation Using N-Body Simulation

- Aim: Simulate formation of Solar System using N-body simulation to study interactions between multiple bodies (ex. Planets, moons, asteroids).
- Analysis: Developed Python script to solve Newton's gravitational equations and simulate evolution of the Solar System over millions of years.
- Results: Different initial conditions such as distribution of mass and velocity affect the final configuration of the system including stable and unstable configurations.

Simulating Stellar Evolution (Hertzsprung-Russell Diagram)

- Aim: Simulate evolution of Sun and plot its position on the HR diagram.
- Analysis: Used stellar evolution equations to model the Sun's life cycle from its formation to its eventual transition to a red giant and white dwarf. Represented its position on the HR diagram, tracking the star's temperature and luminosity over time.
- Results: The Sun follows a predictable life cycle on the HR diagram, moving from the main sequence to the red giant phase, and eventually becoming a white dwarf. Comparing its evolution with other stars on the HR diagram explains the characteristics of stars at different phases in their life cycle.

Simulating Orbital Mechanics

- Aim: Model the motion of two celestial bodies (Earth and moon) based on the gravitational forces.
- Analysis: Used different numerical methods such as Euler's and Runge-Kutta to demonstrate the trajectories of bodies over time. Analyzed how different initial conditions affect the orbital paths.

- Results: Demonstrated the application of Newton's laws of motion and gravitational forces in modeling the orbital dynamics of two celestial bodies. The results showed that small changes in the initial conditions, such as velocity or distance between the two bodies, significantly affect the shape and stability of their orbits.

Study of Nature of Neutrino: Majorana or Dirac

- Aim: Investigating whether neutrinos are Majorana (their own antiparticles) or Dirac particles, with implications for processes like neutrinoless double beta decay.
- Analysis: The study compared the Seesaw Mechanism, effective Majorana mass, and lepton number violation to assess the neutrino's nature, focusing on experimental data and theoretical models.
- Results: The study showed that findings of the neutrinoless double beta decay could point to the Majorana nature of neutrinos. However, further experimental evidence is needed to confirm this theory.

Certifications

- IISC: Winter School on Recent Frontiers in Physics Research: Matter, Life, and Cosmos
- ICTP: Summer School on Cosmology
- IISER: Physics through Computational Thinking
- Internshala: Programming with Python
- Amazon Web Services: Getting started with AWS Machine Learning
- IISER: Waves and Oscillations

Skills

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|----------|----------|---------|-------|
| • Python | • Topcat | • LaTeX | • DS9 |
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Languages

German

Basic Knowledge (A2)

English

Business Fluent (C1)